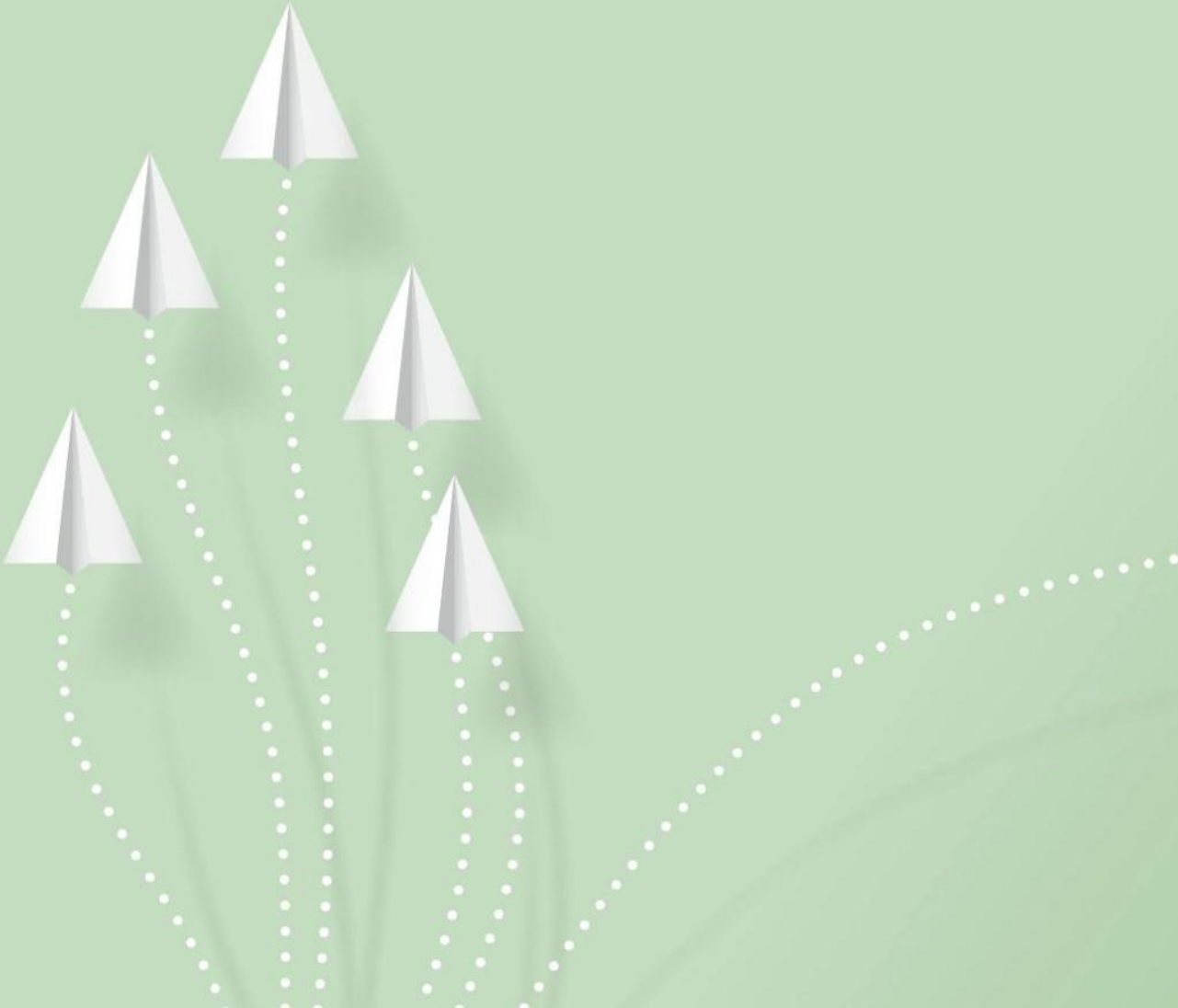




Real Estate Data Analysis Using SQL

EXPLORING PROPERTY TRENDS,
PRICING PATTERNS, AND
INVESTMENT INSIGHTS

Find duplicate properties based on same Location, Purchase Price and Square Footage

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila
 sql_project
 Tables
 real_estate_1
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Administration Schemas

Information

Table: real_estate_1

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Real_Estate_Project_1*

Limit to 1000 rows

```
10  
11  
12 #1. Find duplicate properties based on same Location, Purchase Price and Square Footage.  
13 • select location, count(*) as total_count  
14 from real_estate_1  
15 group by location  
16 having total_count>1;  
17  
18
```

Result Grid

	location	total_count
▶	Houston	129
	San Francisco	124
	Chicago	108
	Los Angeles	115
	New York	127

Result 15 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 23	16:04:58	select location, count(*) as total_count from real_estate_1 group by locati...	5 row(s) returned	0.000 sec / 0.000 sec
✓ 24	16:05:12	select location, count(*) as total_count from real_estate_1 group by locati...	5 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Result Grid
Form Editor
Read Only

Find duplicate properties based on same Location, Purchase Price and Square Footage

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator:

SCHEMAS

Filter objects

sakila

sql_project

Tables

real_estate_1

real_estate_2

Views

Stored Procedures

Functions

sys

world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
16
17 • select `Purchase Price ($)` , count(*) as Total_count
18 from real_estate_1
19 group by `Purchase Price ($)`
20 having Total_count>1;
21
22
23
24
```

Result Grid

	Purchase Price (\$)	Total_count
--	---------------------	-------------

Result 16

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 24	16:05:12	select location, count(*) as total_count from real_estate_1 group by locati...	5 row(s) returned	0.000 sec / 0.000 sec
✓ 25	16:06:37	select `Purchase Price (\$)` , count(*) as Total_count from real_estate_1 gr...	0 row(s) returned	0.000 sec / 0.000 sec

Find duplicate properties based on same Location, Purchase Price and Square Footage

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila

sql_project

Tables

real_estate_1

real_estate_2

Views

Stored Procedures

Functions

sys

world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime double

Rate int

Walkability Score int

Public Transport text

Access int

Final Sale Price (\$) int

Final Sale Price (\$) int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
22
23 • select `Square Footage (sq ft)`, count(*) as total_count
24 from real_estate_1
25 group by `Square Footage (sq ft)`
26 having total_count > 1;
27
28
29
30
```

Result Grid

	Square Footage (sq ft)	total_count
▶		28
	4875	2
	2010	3
	3909	2
	1145	3
	1465	1465
	3803	2

Result 18 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 26	16:07:32	select 'Square Footage (sq ft)', count(*) as total_count from real_estate_1 ...	30 row(s) returned	0.000 sec / 0.000 sec
✓ 27	16:07:42	select 'Square Footage (sq ft)', count(*) as total_count from real_estate_1 ...	30 row(s) returned	0.000 sec / 0.000 sec

. Find the second highest Final Sale Price among all properties.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHMAS

Filter objects

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 real_estate_1
 real_estate_2
 Views
 Stored Procedures
 Functions
 sys
 world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1* x

Limit to 1000 rows

```
25 -----  
26 #2.Find the second highest Final Sale Price among all properties.  
27 • select property_ID, 'Final Sale Price ($)`  
28 from real_estate_2  
29 order by 'Final Sale Price ($)`desc  
30 limit 1 offset 1;  
31  
32  
33
```

Result Grid

	property_ID	Final Sale Price (\$)
▶	182	1198053

real_estate_2 19 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	27 16:07:42	select 'Square Footage (sq ft)', count(*) as total_count from real_estate_1 ...	30 row(s) returned	0.000 sec / 0.000 sec
✓	28 16:09:41	select property_ID, 'Final Sale Price (\$)` from real_estate_2 order by 'Final ...	1 row(s) returned	0.000 sec / 0.000 sec

Result Grid
Form Editor
Read Only

. Retrieve properties that are present in real_estate_1 but missing in real_estate_2.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'sakila' and 'sql_project' databases. Under 'sql_project', the 'real_estate_1' and 'real_estate_2' tables are listed. The 'Information' tab for 'real_estate_2' shows its columns: Property_Id (int), Property Type (text), Year Built (int), Neighborhood Crime Rate (double), Walkability Score (int), Public Transport Access (text), Final Sale Price (\$) (int), Final Sale Price (\$) [0] (int), Number of Bedrooms (int), and Number of Bathrooms (int).

The main editor window shows a SQL query in the 'Real_Estate_Project_1' tab:

```
31  
32 #3. Retrieve properties that are present in real_estate_1 but missing in real_estate_2.  
33 • select * from real_estate_1  
34 left join real_estate_2  
35 on real_estate_1.Property_Id= real_estate_2.property_Id  
36 where real_estate_2.Property_Id is null;  
37  
38  
39
```

The 'Result Grid' tab shows the results of the query. The table has 9 columns: Property_Id, Location, Purchase Price (\$), Annual Rent Income (\$), Maintenance Costs (\$), Loan Interest Rate (%), Loan Term (Years), Property Tax (%), and Appreciation (%). The results are empty.

The 'Output' tab shows the execution log:

#	Time	Action	Message	Duration / Fetch
✓ 28	16:09:41	select property_ID, 'Final Sale Price (\$)' from real_estate_2 order by 'Final ...	1 row(s) returned	0.000 sec / 0.000 sec
✓ 29	16:11:23	select * from real_estate_1 left join real_estate_2 on real_estate_1.Property...	0 row(s) returned	0.016 sec / 0.000 sec

Get the top 5 most expensive properties based on Final Sale Price.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

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 Tables
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 real_estate_2
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 sys
 world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1* x

Limit to 1000 rows

```
#4. Get the top 5 most expensive properties based on Final Sale Price.  
select Property_ID, `Property Type`, `Final Sale Price ($)`  
from real_estate_2  
order by `Final Sale Price ($)` desc  
limit 5;
```

Result Grid

	Property_ID	Property Type	Final Sale Price (\$)
▶	509	Apartment	1198702
	182	Apartment	1198053
	426	Townhouse	1197158
	310	Villa	1195622
	442	Townhouse	1193565

real_estate_2 21 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 29	16:11:23	select * from real_estate_1 left join real_estate_2 on real_estate_1.Property...	0 row(s) returned	0.016 sec / 0.000 sec
✓ 30	16:12:56	select Property_ID, `Property Type`, `Final Sale Price (\$)` from real_estate_...	5 row(s) returned	0.000 sec / 0.000 sec

Show the total number of properties in each city.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila
 sql_project
 Tables
 real_estate_1
 real_estate_2
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 world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
#5.Show the total number of properties in each city.  
select location,count(real_estate_1. Property_ID) as No_Of_Property  
from real_estate_1  
group by location  
order by No_Of_Property;
```

Result Grid

location	No_Of_Property
Chicago	108
Los Angeles	115
San Francisco	124
New York	127
Houston	129

Result 22

Output

Action Output

#	Time	Action	Message	Duration / Fetch
30	16:12:56	select Property_ID,'Property Type','Final Sale Price (\$)' from real_estate_...	5 row(s) returned	0.000 sec / 0.000 sec
31	16:13:50	select location,count(real_estate_1. Property_ID) as No_Of_Property from...	5 row(s) returned	0.000 sec / 0.000 sec

Calculate the average Annual Rent Income for each city.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- sakila
- sql_project
 - Tables
 - real_estate_1
 - real_estate_2
 - Views
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- sys
- world

Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
50
51
52 -----
53 #6. Calculate the average Annual Rent Income for each city.
54 • select location, avg(`Annual Rent Income ($)`) as Avg_Rent
55   from real_estate_1
56   group by Location
57   order by Avg_Rent desc;
58
```

Result Grid

	location	Avg_Rent
▶	New York	35417.6142
	Houston	35237.1085
	Los Angeles	35159.1826
	Chicago	34158.3056
	San Francisco	33689.3790

Result 23 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 31	16:13:50	select location,count(real_estate_1. Property_ID) as No_Of_Property from...	5 row(s) returned	0.000 sec / 0.000 sec
✓ 32	16:15:14	select location, avg(`Annual Rent Income (\$)`) as Avg_Rent from real_est...	5 row(s) returned	0.016 sec / 0.000 sec

Find cities where average Final Sale Price is higher than the overall average Final Sale Price.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator: Real_Estate_Project_1*

SCHEMAS

Filter objects

- sakila
- sql_project
 - Tables
 - real_estate_1
 - real_estate_2
 - Views
 - Stored Procedures
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- sys
- world

Administration Schemas

Information

Columns:

- Property_Id int
- Property Type text
- Year Built int
- Neighborhood Crime Rate double
- Walkability Score int
- Public Transport Access text
- Final Sale Price (\$) int
- Final Sale Price (\$) [0] int
- Number of Bedrooms int
- Number of Bathrooms int

Result Grid

	location	Avg_Price
▶	Chicago	675706.7685
	New York	708563.0079

Result 24 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 32	16:15:14	select location, avg('Annual Rent Income (\$)') as Avg_Rent from real_est...	5 row(s) returned	0.016 sec / 0.000 sec
✓ 33	16:16:42	select location, avg('Final Sale Price (\$)') as Avg_Price from real_estate_1...	2 row(s) returned	0.015 sec / 0.000 sec

```
#7. Find cities where average Final Sale Price is higher than the overall average Final Sale Price.
select location, avg('Final Sale Price ($)') as Avg_Price
from real_estate_1
join real_estate_2
on real_estate_1.property_id=real_estate_2.property_id
group by location
having avg('Final Sale Price ($)') >
(select avg('Final Sale Price ($)') from real_estate_2);
```

Retrieve properties whose Final Sale Price is above the overall average Final Sale Price.

The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'sakila' and 'sql_project' databases. The 'sql_project' database is expanded, showing tables 'real_estate_1' and 'real_estate_2'. The main editor window shows a SQL query for 'Real_Estate_Project_1*':

```
#8.Retrieve properties whose Final Sale Price is above the overall average Final Sale Price.
select real_estate_1.Property_Id,Location,`Final Sale Price ($)`
from real_estate_1
join real_estate_2
on real_estate_1.Property_Id=real_estate_2.property_Id
where `Final Sale Price ($)`>
(select avg(`Final Sale Price ($)` ) from real_estate_2);
```

The 'Result Grid' shows the following data:

	Property_Id	Location	Final Sale Price (\$)
▶	1	Houston	720086
	3	Chicago	765275
	6	Los Angeles	1129293
	8	Chicago	732049
	12	Chicago	1118057
	14	Los Angeles	1151497
	15	Houston	1022180

The 'Information' panel on the left lists the columns for 'real_estate_2':

Columns:	
Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

The 'Output' panel at the bottom shows the execution results:

#	Time	Action	Message	Duration / Fetch
✓ 33	16:16:42	select location, avg('Final Sale Price (\$)') as Avg_Price from real_estate_1...	2 row(s) returned	0.015 sec / 0.000 sec
✓ 34	16:17:56	select real_estate_1.Property_Id,Location,`Final Sale Price (\$)` from real_e...	306 row(s) returned	0.000 sec / 0.000 sec

Find the most expensive property in each city.

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila

sql_project

Tables

real_estate_1

real_estate_2

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Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) [0] int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
73 #9.Find the most expensive property in each city.
74 select * from (select real_estate_1.property_Id,location,'Final Sale Price ($)',
75 rank() over (partition by location
76 order by 'Final Sale Price ($)' desc) as Price_Rank
77 from real_estate_1
78 join real_estate_2
79 on real_estate_1.property_Id=real_estate_2.Property_Id) ranked_properties
80 where Price_Rank=1;
81
```

Result Grid

	property_Id	location	Final Sale Price (\$)	Price_Rank
▶	182	Chicago	1198053	1
	267	Houston	1170572	1
	426	Los Angeles	1197158	1
	611	New York	1191838	1
	310	San Francisco	1195622	1

Result 26 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 34	16:17:56	select real_estate_1.Property_Id,Location,'Final Sale Price (\$)' from real_e...	306 row(s) returned	0.000 sec / 0.000 sec
✓ 35	16:19:09	select * from (select real_estate_1.property_Id,location,'Final Sale Price (\$...	5 row(s) returned	0.015 sec / 0.000 sec

Find the cheapest property in each city.

MySQL Workbench

Local instance MySQL80

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Navigator

SCHEMAS

Filter objects

sakila

sql_project

Tables

real_estate_1

real_estate_2

Views

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Functions

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world

Administration Schemas

Information

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) [0] int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
83
84 #10.Find the cheapest property in each city.
85 select * from (select real_estate_1.Property_Id,location,`Final Sale Price ($)`,
86 rank() over(partition by Location order by `Final Sale Price ($)` asc) as Rank_By_Price
87 from real_estate_1
88 join real_estate_2
89 on real_estate_1.property_Id=real_estate_2.property_Id) as ranked_Prices
90 where Rank_by_Price=1;
91
```

Result Grid

	Property_Id	location	Final Sale Price (\$)	Rank_By_Price
▶	136	Chicago	115909	1
	697	Houston	109387	1
	104	Los Angeles	101832	1
	272	New York	108888	1
	445	San Francisco	101379	1

Result 27 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 35	16:19:09	select * from (select real_estate_1.property_Id,location,`Final Sale Price (\$...	5 row(s) returned	0.015 sec / 0.000 sec
✓ 36	16:20:54	select * from (select real_estate_1.Property_Id,location,`Final Sale Price (\$...	5 row(s) returned	0.015 sec / 0.000 sec

Rank properties by Final Sale Price within each city.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- sakila
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- sys
- world

Administration Schemas

Information

Columns:

- Property_Id int
- Property Type text
- Year Built int
- Neighborhood Crime Rate double
- Walkability Score int
- Public Transport Access text
- Final Sale Price (\$) int
- Final Sale Price (\$)_[0] int
- Number of Bedrooms int
- Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
92 -----
93 #11.Rank properties by Final Sale Price within each city.
94 select * from (select real_estate_1.property_Id, Location,`Final Sale Price ($)`,
95 rank() over(partition by location order by `Final Sale Price ($)`) as Rank_By_Price
96 from real_estate_1
97 join real_estate_2
98 on real_estate_1.property_Id=real_estate_2.Property_Id)
99 Ranked_properties;
```

100

Result Grid

	property_Id	Location	Final Sale Price (\$)	Rank_By_Price
▶	136	Chicago	115909	1
	254	Chicago	127690	2
	135	Chicago	136051	3
	109	Chicago	137570	4
	202	Chicago	161009	5
	447	Chicago	169306	6
	206	Chicago	171942	7

Result 28 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 36	16:20:54	select * from (select real_estate_1.Property_Id,location,`Final Sale Price (\$...	5 row(s) returned	0.015 sec / 0.000 sec
✓ 37	16:22:58	select * from (select real_estate_1.property_Id, Location,`Final Sale Price (...	603 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Result Grid

Form Editor

Read Only

Calculate the difference between highest and lowest Final Sale Price in each city.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator:

SCHMAS

Filter objects

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- sql_project
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 - real_estate_1
 - real_estate_2
 - Views
 - Stored Procedures
 - Functions
- sys
- world

Administration Schemas

Information

Columns:

- Property_Id int
- Property Type text
- Year Built int
- Neighborhood Crime double
- Rate int
- Walkability Score int
- Public Transport Access text
- Final Sale Price (\$) int
- Final Sale Price (\$) [0] int
- Number of Bedrooms int
- Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
98 -----
99 #12.Calculate the difference between highest and lowest Final Sale Price in each city.
100 • select location,max(`Final Sale Price ($)`,`),min(`Final Sale Price ($)`,`),
101      max(`Final Sale Price ($)`,`)-min(`Final Sale Price ($)`,`) as Difference
102 from real_estate_1
103 join real_estate_2
104 on real_estate_1.property_Id=real_estate_2.property_Id
105 group by location;
106
```

Result Grid

	location	max(`Final Sale Price (\$)`,`)	min(`Final Sale Price (\$)`,`)	Difference
▶	Houston	1170572	109387	1061185
	San Francisco	1195622	101379	1094243
	Chicago	1198053	115909	1082144
	Los Angeles	1197158	101832	1095326
	New York	1191838	108888	1082950

Result 29 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 37	16:22:58	select * from (select real_estate_1.property_Id, Location,'Final Sale Price (...)	603 row(s) returned	0.016 sec / 0.000 sec
✓ 38	16:25:15	select location,max(`Final Sale Price (\$)`,`),min(`Final Sale Price (\$)`,`), max(`...	5 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Result Grid

Form Editor

Read Only

Show the percentage contribution of each city to total Final Sale value of all properties.

MySQL Workbench

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

sakila

sql_project

Tables

real_estate_1

real_estate_2

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Administration Schemas

Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) [0] int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
107 #13.Show the percentage contribution of each city to total Final Sale value of all properties.
108 • select location, sum(`Final Sale Price ($)`) as City_Total_Sales,
109 round(sum(`Final Sale Price ($)`) * 100 / sum(sum(`Final Sale Price ($)`) over(), 2) as percentage_contribution
110 from real_estate_1
111 join real_estate_2
112 on real_estate_1.Property_Id=real_estate_2.Property_Id
113 group by location
114 order by City_Total_Sales desc;
115
```

Result Grid

	location	City_Total_Sales	percentage_contribution
▶	New York	89987502	22.72
	Houston	82859288	20.92
	San Francisco	79467001	20.06
	Chicago	72976331	18.42
	Los Angeles	70798691	17.87

Result 30 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 38	16:25:15	select location,max(`Final Sale Price (\$)`,min(`Final Sale Price (\$)`, max(`...	5 row(s) returned	0.016 sec / 0.000 sec
✓ 39	16:26:38	select location, sum(`Final Sale Price (\$)`) as City_Total_Sales, round(sum...	5 row(s) returned	0.016 sec / 0.000 sec

Object Info Session

Result Grid

Form Editor

Read Only

Find properties where Final Sale Price is higher than Purchase Price.

The screenshot shows the MySQL Workbench interface with a query window titled 'Real_Estate_Project_1'. The query is as follows:

```
#14.Find properties where Final Sale Price is higher than Purchase Price.
select real_estate_1.property_Id,`Purchase Price ($)``,`Final Sale Price ($)`
from real_estate_1
join real_estate_2
on real_estate_1.property_Id=real_estate_2.property_Id
where real_estate_2.`Final Sale Price ($)`> real_estate_1.`Purchase Price ($)`;
```

The 'Result Grid' shows the following data:

	property_Id	Purchase Price (\$)	Final Sale Price (\$)
▶	1	686165	720086
	3	412066	765275
	6	661419	1129293
	12	399770	1118057
	14	581680	1151497
	15	381342	1022180
	17	144238	1118225

The 'Information' panel on the left lists the columns for the 'real_estate_1' table:

- Property_Id: int
- Property Type: text
- Year Built: int
- Neighborhood Crime: double
- Rate: double
- Walkability Score: int
- Public Transport Access: text
- Final Sale Price (\$): int
- Final Sale Price (\$)_[0]: int
- Number of Bedrooms: int
- Number of Bathrooms: int

The 'Output' panel at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
39	16:26:38	select location, sum('Final Sale Price (\$)') as City_Total_Sales, round(sum...	5 row(s) returned	0.016 sec / 0.000 sec
40	16:27:52	select real_estate_1.property_Id,`Purchase Price (\$)``,`Final Sale Price (\$)` ...	364 row(s) returned	0.000 sec / 0.000 sec

Calculate price per square foot for each property.

MySQL Workbench

Local instance MySQL80 x

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Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
122 -----  
123 #15. Calculate price per square foot for each property.  
124 • select real_estate_1.property_Id,`Square Footage (sq ft)`,`Final Sale Price ($)`,  
125 round(`Final Sale Price ($)`,`Square Footage (sq ft)`),2) as Price_Per_Square_Foot  
126 from real_estate_1  
127 join real_estate_2  
128 on real_estate_1.property_Id=real_estate_2.property_Id;  
129  
130
```

Result Grid

	property_Id	Square Footage (sq ft)	Final Sale Price (\$)	Price_Per_Square_Foot
▶	1	3876	720086	185.78
	2	1899	259876	136.85
	3		765275	NULL
	4	2135	305229	142.96
	5	677	110547	163.29
	6	3431	1129293	329.14
	8	3321	732049	220.43

Result 32 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 40	16:27:52	select real_estate_1.property_Id,`Purchase Price (\$)`,`Final Sale Price (\$)` ...	364 row(s) returned	0.000 sec / 0.000 sec
✓ 41	16:29:10	select real_estate_1.property_Id,`Square Footage (sq ft)`,`Final Sale Price` ...	603 row(s) returned	0.016 sec / 0.000 sec

Find properties whose price per square foot is higher than their city's average price per square foot.

The screenshot shows the MySQL Workbench interface with a query window titled 'Real_Estate_Project_1'. The query is as follows:

```
131 -----
132 #16.Find properties whose price per square foot is higher than their city's average price per square foot.
133 • select * from (select real_estate_1.property_Id,('Final Sale Price ($)'/'Square Footage (sq ft)') as price_per_sq_feet,
134 avg('Final Sale Price ($)'/'Square Footage (sq ft)') over (partition by location) as city_avg_price
135 from real_estate_1
136 join real_estate_2
137 on real_estate_1.Property_Id=real_estate_2.Property_Id) as T
138 where price_per_sq_feet>city_avg_price;
139
```

The 'Result Grid' shows the following data:

property_Id	price_per_sq_feet	city_avg_price
444	483.51988279614903	344.676933196826
331	879.7836257309941	344.676933196826
443	511.72597526165555	344.676933196826
12	430.68451463790444	344.676933196826
685	1498.6604278074867	344.676933196826
398	554.1649061032864	344.676933196826
678	387.0990675990676	344.676933196826
131	1106.3588110403398	344.676933196826

The 'Object Info' pane on the left shows the schema for 'real_estate_2' with columns: Property_Id (int), Property Type (text), Year Built (int), Neighborhood Crime Rate (double), Walkability Score (int), Public Transport Access (text), Final Sale Price (\$) (int), Final Sale Price (\$)_[0] (int), Number of Bedrooms (int), and Number of Bathrooms (int).

The 'Output' pane at the bottom shows the execution log:

#	Time	Action	Message	Duration / Fetch
✓ 41	16:29:10	select real_estate_1.property_Id,'Square Footage (sq ft)','Final Sale Price ...	603 row(s) returned	0.016 sec / 0.000 sec
✓ 42	16:30:45	select * from (select real_estate_1.property_Id,('Final Sale Price (\$)'/'Squa...	209 row(s) returned	0.016 sec / 0.000 sec

.Calculate the average Final Sale Price for each Property Type.

MySQL Workbench

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Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) [0] int

Number of Bedrooms int

Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
137
138
139 -----
140 #17.Calculate the average Final Sale Price for each Property Type.
141 • select `Property Type`, avg(`Final Sale Price ($)`) as avg_sale_price_per_city
142 from real_estate_2
143 group by `Property Type`
144 order by avg_sale_price_per_city desc;
145
```

Result Grid

	Property Type	avg_sale_price_per_city
▶	Townhouse	713513.3571
	Condo	660309.0526
	Villa	659374.6020
	Apartment	633798.1229

Result 34 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓	42 16:30:45	select * from (select real_estate_1.property_Id,('Final Sale Price (\$)')/'Squa...	209 row(s) returned	0.016 sec / 0.000 sec
✓	43 16:31:57	select `Property Type`, avg(`Final Sale Price (\$)`) as avg_sale_price_per_...	4 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Result Grid

Form Editor

Read Only

Find the Property Type with the highest average Final Sale Price.

MySQL Workbench

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Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$)_[0] int

Number of Bedrooms int

Number of Bathrooms int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
143 -----
144 #18.Find the Property Type with the highest average Final Sale Price.
145 • select `Property Type`, avg(`Final Sale Price ($)`) as avg_sale_price_per_city
146 from real_estate_2
147 group by `Property Type`
148 order by avg_sale_price_per_city desc
149 limit 1;
150
151
```

Result Grid

Property Type	avg_sale_price_per_city
Townhouse	713513.3571

Result 36

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 44	16:33:13	select 'Property Type', avg('Final Sale Price (\$)') as avg_sale_price_per_...	4 row(s) returned	0.015 sec / 0.000 sec
✓ 45	16:33:38	select 'Property Type', avg('Final Sale Price (\$)') as avg_sale_price_per_...	1 row(s) returned	0.000 sec / 0.000 sec

Count the number of properties for each Property Type.

MySQL Workbench

Local instance MySQL80 x

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Information

Columns:

- Property_Id int
- Property Type text
- Year Built int
- Neighborhood Crime double
- Rate int
- Walkability Score int
- Public Transport text
- Access int
- Final Sale Price (\$) int
- Final Sale Price int
- (\$)_[0] int
- Number of Bedrooms int
- Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
150 -----
151 #19.Count the number of properties for each Property Type.
152 • select `Property Type`,count(Property_Id) as total_count
153 from real_estate_2
154 group by `Property Type`
155 order by total_count desc;
156
157
158
```

Result Grid

Property Type	total_count
Villa	196
Apartment	179
Condo	171
Townhouse	154

Result 37 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
45	16:33:38	select 'Property Type', avg('Final Sale Price (\$)') as avg_sale_price_per_...	1 row(s) returned	0.000 sec / 0.000 sec
46	16:35:59	select 'Property Type',count(Property_Id) as total_count from real_estate_...	4 row(s) returned	0.000 sec / 0.000 sec

Find properties whose Vacancy Rate is higher than the average Vacancy Rate of their city.

MySQL Workbench

Local instance MySQL80 x

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Information

Table: real_estate_2

Columns:

Property_Id int
Property Type text
Year Built int
Neighborhood Crime Rate double
Walkability Score int
Public Transport Access text
Final Sale Price (\$) int
Final Sale Price (\$) [0] int
Number of Bedrooms int
Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
156  
157 #20. Find properties whose Vacancy Rate is higher than the average Vacancy Rate of their city.  
158 • select * from (select Property_ID, location, `Vacancy Rate (%)`,  
159 avg (`Vacancy Rate (%)`) over (partition by Location) as City_Avg_Vacancy_Rate from real_estate_1) as A  
160 where `Vacancy Rate (%)` > City_Avg_Vacancy_Rate;  
161  
162  
163  
164
```

Result Grid

	Property_ID	location	Vacancy Rate (%)	City_Avg_Vacancy_Rate
▶	331	Chicago	9.03	5.171851851851851
	443	Chicago	8.62	5.171851851851851
	12	Chicago	5.93	5.171851851851851
	398	Chicago	8.62	5.171851851851851
	678	Chicago	6.62	5.171851851851851
	177	Chicago	6.52	5.171851851851851
	660	Chicago	7.25	5.171851851851851

Result 38 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 46	16:35:59	select 'Property Type', count(Property_Id) as total_count from real_estate_...	4 row(s) returned	0.000 sec / 0.000 sec
✓ 47	16:36:46	select * from (select Property_ID, location, `Vacancy Rate (%)`, avg (`Vaca...	309 row(s) returned	0.000 sec / 0.000 sec

Find properties built after the average Year Built of all properties.

MySQL Workbench

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Information

Columns:

Property_Id int
Property Type text
Year Built int
Neighborhood Crime Rate double
Walkability Score int
Public Transport Access text
Final Sale Price (\$) int
Final Sale Price (\$) [0] int
Number of Bedrooms int
Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
#21. Find properties built after the average Year Built of all properties.  
select * from real_estate_2  
where `Year Built` >  
(select avg(`Year built`) from real_estate_2);
```

Result Grid

	Property_Id	Property Type	Year Built	Neighborhood Crime Rate	Walkability Score	Public Transport Access	Final Sale Price (\$)	Final Sale Price (\$) [0]	Number Bedroom
▶	1	Villa	1995	24.46	20	No	720086	720086	2
	2	Apartment	2022	19.15	85	No	259876	259876	1
	3	Villa	2006	11.1	70	No	765275	765275	5
	5	Condo	2004	30.42	22	Yes	110547	110547	2
	6	Apartment	2010	22.22	21	No	1129293	1129293	2
	9	Townhouse	1997	42.51	29	No	182970	182970	4

real_estate_2 39 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 47	16:36:46	select * from (select Property_ID, location, 'Vacancy Rate (%)', avg ('Vaca...	309 row(s) returned	0.000 sec / 0.000 sec
✓ 48	16:38:22	select * from real_estate_2 where `Year Built` > (select avg(`Year built`) fro...	348 row(s) returned	0.016 sec / 0.000 sec

Find the top 3 cities by total Final Sale value.

MySQL Workbench

Local instance MySQL80 x

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Information

Table: real_estate_2

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1* x

Limit to 1000 rows

```
#23. Find the top 3 cities by total Final Sale value.  
select location, sum(`Final Sale Price ($)`) as total_Final_Sale  
from real_estate_1  
join real_estate_2  
on real_estate_1.property_Id=real_estate_2.property_id  
group by location  
order by total_final_sale desc  
limit 3;
```

Result Grid

	location	total_Final_Sale
▶	New York	89987502
	Houston	82859288
	San Francisco	79467001

Result 40 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 48	16:38:22	select * from real_estate_2 where `Year Built` > (select avg(`Year built`) fro...	348 row(s) returned	0.016 sec / 0.000 sec
✓ 49	16:39:45	select location, sum(`Final Sale Price (\$)`) as total_Final_Sale from real_est...	3 row(s) returned	0.000 sec / 0.000 sec

Compare average Final Sale Price of properties with and without Public Transport access.

MySQL Workbench

Local instance MySQL80 x

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Information

Columns:

Property_Id	int
Property Type	text
Year Built	int
Neighborhood Crime Rate	double
Walkability Score	int
Public Transport Access	text
Final Sale Price (\$)	int
Final Sale Price (\$)_[0]	int
Number of Bedrooms	int
Number of Bathrooms	int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
181 -----
182 #26. Compare average Final Sale Price of properties with and without Public Transport access.
183 • Select `Public Transport Access`, AVG(`Final Sale Price ($)`) AS avg_final_sale_price
184 FROM real_estate_2
185 GROUP BY `Public Transport Access`;
```

Result Grid

	Public Transport Access	avg_final_sale_price
▶	No	647394.7575
	Yes	684346.2913

Result 41 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 49	16:39:45	select location,sum(`Final Sale Price (\$)`) as total_Final_Sale from real_est...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 50	16:40:48	Select `Public Transport Access`, AVG(`Final Sale Price (\$)`) AS avg_final...	2 row(s) returned	0.015 sec / 0.000 sec

Find properties where Crime Rate is lower than the average Crime Rate of their city.

MySQL Workbench

Local instance MySQL80 x

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Information

Columns:

Property_Id int
Property Type text
Year Built int
Neighborhood Crime Rate double
Walkability Score int
Public Transport Access text
Final Sale Price (\$) int
Final Sale Price (\$) [0] int
Number of Bedrooms int
Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
188 -----  
189 #27. Find properties where Crime Rate is lower than the average Crime Rate of their city.  
190 • select real_estate_1.property_Id,location,'Neighborhood Crime Rate'  
191 from real_estate_1  
192 join real_estate_2  
193 on real_estate_1.property_id=real_estate_2.property_Id  
194 where 'Neighborhood Crime Rate'< (select avg('Neighborhood Crime Rate') from real_estate_2);  
195  
196
```

Result Grid

	property_Id	location	Neighborhood Crime Rate
▶	1	Houston	24.46
	2	San Francisco	19.15
	3	Chicago	11.1
	6	Los Angeles	22.22
	8	Chicago	2.39
	13	San Francisco	18.72
	14	Los Angeles	11.57

Result 42 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 50	16:40:48	Select 'Public Transport Access', AVG('Final Sale Price (\$)') AS avg_final...	2 row(s) returned	0.015 sec / 0.000 sec
✓ 51	16:41:49	select real_estate_1.property_Id,location,'Neighborhood Crime Rate' from ...	296 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

Rank properties by Walkability Score within each city.

MySQL Workbench

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Information

Table: real_estate_2

Columns:

Property_Id int

Property Type text

Year Built int

Neighborhood Crime Rate double

Walkability Score int

Public Transport Access text

Final Sale Price (\$) int

Final Sale Price (\$) [0] int

Number of Bedrooms int

Number of Bathrooms int

Object Info Session

Real_Estate_Project_1*

Limit to 1000 rows

```
#28.Rank properties by Walkability Score within each city.
select real_estate_1.Property_Id,Location,'Walkability Score',
rank() over(partition by Location order by 'Walkability Score') as Score_By_Walkability
from real_estate_1
join real_estate_2
on real_estate_1.property_Id= real_estate_2.property_Id;
```

Result Grid

	Property_Id	Location	Walkability Score	Score_By_Walkability
▶	616	Chicago	11	1
	213	Chicago	12	2
	401	Chicago	13	3
	114	Chicago	13	3
	197	Chicago	14	5
	26	Chicago	14	5
	168	Chicago	15	7

Result 43 x

Read Only

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 51	16:41:49	select real_estate_1.property_Id,Location,'Neighborhood Crime Rate' from ...	296 row(s) returned	0.000 sec / 0.000 sec
✓ 52	16:43:27	select real_estate_1.Property_Id,Location,'Walkability Score', rank() over(...	603 row(s) returned	0.000 sec / 0.000 sec

Find the property in each city with the maximum difference between Purchase Price and Final Sale Price.

MySQL Workbench

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Information

Columns:

Property_Id int
Property Type text
Year Built int
Neighborhood Crime double
Rate double
Walkability Score int
Public Transport text
Access text
Final Sale Price (\$) int
Final Sale Price (\$) int
Number of Bedrooms int
Number of Bathrooms int

Real_Estate_Project_1*

Limit to 1000 rows

```
200 -----  
201 #29. Find the property in each city with the maximum difference between Purchase Price and Final Sale Price.  
202 select * from (select real_estate_1.Property_Id,location, `Purchase Price ($)`-`Final Sale Price ($)` as Difference,  
203 rank() over(partition by location order by `Purchase Price ($)`-`Final Sale Price ($)`desc) as Difference_rank from  
204 real_estate_1  
205 join real_estate_2  
206 on real_estate_1.property_id=real_estate_2.property_id) as A  
207 where Difference_Rank=1;  
208 -----
```

Result Grid

	Property_Id	location	Difference	Difference_rank
▶	136	Chicago	656199	1
	80	Houston	877546	1
	58	Los Angeles	843196	1
	248	New York	751953	1
	5	San Francisco	676862	1

Result 44 x

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 52	16:43:27	select real_estate_1.Property_Id,Location,`Walkability Score`,rank() over(...	603 row(s) returned	0.000 sec / 0.000 sec
✓ 53	16:44:25	select * from (select real_estate_1.Property_Id,location,`Purchase Price (...	5 row(s) returned	0.000 sec / 0.000 sec